

Smart Gas Detection and Environment Monitoring System

Components Required

- Arduino UNO
- DHT22 Temperature & Humidity Sensor
- MQ2 Gas Sensor
- 16x2 I2C LCD Display
- Red LED
- Jumper Wires
- USB Cable / Power Supply
- Breadboard (Optional)

Pin Connections

Component Pin	Arduino UNO
DHT22 VCC	5V
DHT22 GND	GND
DHT22 DATA	D7
MQ2 VCC	5V
MQ2 GND	GND
MQ2 AO	A0
LCD VCC	5V
LCD GND	GND
LCD SDA	A4
LCD SCL	A5
LED +	D8
LED -	GND

Working Principle

The Arduino UNO reads temperature and humidity from the DHT22 sensor and gas concentration from the MQ2 sensor. The values are displayed on the LCD. If the gas level is high, the LCD displays 'Gas: DANGER' and the red LED turns ON; otherwise it displays 'Gas: SAFE' and the LED remains OFF.

Project Link

<https://wokwi.com/projects/468286161590235137>